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BMJ Open

Skill decay following Basic Life Support training – a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051959
Article Type:	Protocol
Date Submitted by the Author:	04-Apr-2021
Complete List of Authors:	Stanley, Benjamin; University of Birmingham College of Medical and Dental Sciences Burton, Thomas; University of Birmingham College of Medical and Dental Sciences Percival, Harriet; University of Birmingham College of Medical and Dental Sciences Beesley, Emily; University of Birmingham College of Medical and Dental Sciences Coffin, Nicholas; University of Birmingham College of Medical and Dental Sciences; Salford Royal NHS Foundation Trust Hulme, Jonathan; University of Birmingham College of Medical and Dental Sciences; Sandwell and West Birmingham Hospitals NHS Trust Owen, Andrew; University of Birmingham College of Medical and Dental Sciences; University Hospitals Birmingham NHS Foundation Trust Alderman, Joe; University of Birmingham College of Medical and Dental Sciences; University Hospitals Birmingham NHS Foundation Trust
Keywords:	MEDICAL EDUCATION & TRAINING, Adult intensive & critical care < ANAESTHETICS, PUBLIC HEALTH

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Skill decay following Basic Life Support training – a systematic review protocol

Word count: 1747

Protocol author: Stanley BT
5th year medical student, University of Birmingham.
Committee member RMD Birmingham

Corresponding author:

Joe Alderman

joseph.alderman@nhs.net

University Department of Anaesthesia & Critical Care

Office 1, Ground Floor, East Block

Heritage Building (Queen Elizabeth Hospital)

Mindelsohn Way

Edgbaston

Birmingham

B15 2TH

Reviewing team

Name	Affiliation	Contact details	Role & tasks
Benjamin Stanley	5th year medical student. University of Birmingham (currently intercalating)	bts692@student.bham.ac.uk	Primary reviewer Student lead Project design, protocol authorship, project delivery, manuscript authorship
Thomas Burton	5th year medical student. University of Birmingham (currently intercalating)	txb619@student.bham.ac.uk	Reviewer Project delivery, manuscript authorship
Harriet Percival	5th year medical student. University of Birmingham	hxp644@student.bham.ac.uk	Reviewer Project delivery, manuscript authorship
Emily Beesley	4th year medical student.	elb712@student.bham.ac.uk	Reviewer Project delivery, manuscript authorship

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	University of Birmingham (currently intercalating)		
Nicholas Coffin	Clinical Fellow in Intensive Care Medicine Salford Royal NHS Foundation Trust	nicholasjames.coffin@nhs.net	Project design, protocol authorship, project delivery, manuscript authorship, project supervision
Jonathan Hulme	Consultant, Intensive Care Medicine & Anaesthesia Sandwell and West Birmingham NHS Trust	jonathan.hulme@nhs.net	Course Director, Resuscitation for Medical Disciplines. Project supervision, manuscript authorship
Andrew Owen	NIHR Academic Clinical Lecturer in Intensive Care Medicine. University of Birmingham	andrew.owen@doctors.org.uk	Senior project supervisor Project design, protocol authorship, project delivery, manuscript authorship, project supervision
Joe Alderman	Anaesthesia & Intensive Care Medicine StR University Hospitals Birmingham NHS Foundation Trust	joseph.alderman@nhs.net	Tie-breaker reviewer Junior project supervisor Project design, protocol authorship, project delivery, manuscript authorship, project supervision

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Title: Skill decay following Basic Life Support training – a systematic review protocol

Keywords:

- Basic Life Support
- Out of Hospital Cardiac Arrest (OHCA)
- Skill Decay
- Refresher Training
- Medical Education

Abstract

Introduction

Survival from Out of Hospital Cardiac Arrest (OHCA) is lower in the United Kingdom (UK) than in several developed nations. Bystander Cardiopulmonary Resuscitation (CPR) is associated with increased rates of survival to hospital discharge following OHCA, prompting the introduction of several initiatives by the UK government to increase rates of bystander CPR, including the inclusion of Basic Life Support (BLS) teaching within the national curriculum. Whilst there is clear benefit in this, increasing evidence suggests poor retention of skills following BLS teaching. The aim of this systematic review is to summarise the literature regarding skill decay following BLS training, reporting particularly the time period over which this occurs, and which components of would-be rescuers' performance of the BLS algorithm are most affected.

Methods and Analysis

A search will be conducted to identify studies in which individuals have received BLS training and received subsequent assessment of their skills at a later date. A search strategy comprising relevant MeSH terms and keywords has been devised with librarian input. Relevant databases will be searched with titles, abstract and full text review conducted independently by two reviewers. Data will be extracted from included studies by two reviewers, with meta-analysis conducted if the appropriate pre-conditions (such as limited heterogeneity) are met.

Ethics and Dissemination

No formal ethical approval is required for this systematic review. Results will be disseminated in the form of manuscript submission to a relevant journal and presentation at relevant meetings. To maximise the public's access to this review's findings, any scientific report will be accompanied by a lay summary posted via social media channels, and a press release disseminated to national and international news agencies.

Registration details

This systematic review was submitted to PROSPERO on 14/02/2021 and was accepted on 17/02/2021. The PROSPERO registration number of this study is CRD42021237233

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Strengths and Limitations of this study

- The results of this review will provide clearer guidance on the optimal management of skill decay following Basic Life Support teaching, enabling more effective 'refresher training' to be implemented
- Support from the University of Birmingham library team has been utilised in optimisation of the search strategy, to ensure all relevant studies may be identified for inclusion
- The study methodology is robust, with clear protocol ensuring each title, abstract and full text is assessed by at least 2 reviewers
- Primary data underlying this study is likely to be heterogenous in nature, leading to challenges with conducting meta-analysis

Introduction

In the United Kingdom (UK) approximately 7-8% of patients who suffer out of hospital cardiac arrest (OHCA) survive to hospital discharge. (1) This is a considerably lower rate of survival compared with several other developed countries. (2,3) A recent report from the National Confidential Enquiry into Patient Outcome and Death (NCEPOD) demonstrates that in cases where bystander Cardiopulmonary Resuscitation (CPR) is commenced, survival to hospital discharge is increased. (4) Overall however, rates of bystander CPR remain low in the UK. (1) Increased availability of bystander CPR for patients in the UK has been set as a priority by the Department of Health to increase survival following OHCA. (5)

In order to accomplish this, the need for increased training in Basic Life Support (BLS) skill has been recognised, with BLS and first aid training a compulsory part of the English school curriculum since 2020. (7) Alongside this mandated training there have also been several national and international initiatives aimed at increasing knowledge and skills in BLS, such as the international 'Restart a Heart Day' - a joint initiative of the UK Resuscitation Council, the British Heart Foundation, British Red Cross and St John's Ambulance. (7)

The Problem:

Whilst there is undoubtedly value in increasing the number of people trained in BLS, there is growing evidence that retention of these skills is poor. Evidence from Randomised Control Trials assessing skill retention in lay populations following BLS training demonstrates a considerable decrease in BLS skills, commencing as early as 6 months following initial training (8,9) In addition there is evidence of poor skill retention in junior doctors (10), medical students (11) and nurses (12). This is despite the ability to adequately perform BLS being a requirement of the General Medical Council's (GMC) 'Outcomes for Graduates'. (13)

Need for this study:

As outlined above, there is growing evidence demonstrating poor retention of BLS skills over time amongst both lay people and healthcare students & professionals. As yet a comprehensive systematic review of this evidence has not been conducted. There is a need for such a review to be carried out to ascertain if and how skills decay, and over what time period this occurs. Such information will facilitate future research on how to best refresh BLS skills. In the long-term, this should inform how

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improvements to refresher training programmes can best be implemented, with the overarching aim of improving rates of bystander CPR, and hence survival, from OHCA.

Methods and analysis

Research Questions:

- Is there evidence to support the hypothesis that Basic Life Support skills decay over time?
  - If there is evidence for skill decay, over what time period does this occur?
  - If there is evidence for skill decay, is there a difference in the way this manifests across different aspects of the BLS algorithm?
- What interventions have been implemented to address skill decay?
  - Is there evidence that these interventions are effective at addressing skill decay?

Study Inclusion Criteria:

- Study type: Systematic reviews, randomised controlled trials (RCTs), observational studies (cross-sectional, case-control or cohort), consensus statements, case series, conference papers and proceedings, other reports.
- Population: Individuals who a) have received training in BLS, and b) whose skill retention has been assessed via any means. Includes:
  - Lay individuals
  - Healthcare professionals
  - Healthcare students
- For randomised controlled trials:
  - Intervention: refresher BLS training delivered via any method
  - Outcome: BLS performance assessed via any method

Study Exclusion Criteria:

- Non-English language studies where translation is not possible or cost effective
- Low quality studies will be excluded from the review following assessment using Critical Appraisal Skills Programme (CASP) tool (14) and critical appraisal by at least two reviewers.
- Studies with severe methodological flaws such that results of the study are likely to be flawed

Search Strategy

Published literature will be found through appropriate medical data sources, searched using relevant MeSH terms. Experienced professional colleagues within the University of Birmingham library services team have been consulted to optimise search terms and execution of the search, allowing all applicable articles and abstracts to be identified. The final search strategy is attached as Appendix 1. The reference lists of included articles will be analysed to allow any further potentially relevant articles to be found. These will then be assessed according to the inclusion and exclusion criteria above and included in the review if appropriate. Additionally, experts within the field of resuscitation research will be contacted regarding any unpublished work which may be relevant. If so, permission will be sought to access this so it may be included where appropriate.

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Data sources

1. Pubmed
2. Medline
3. Embase
4. CINAHL
5. Google scholar
6. NICE evidence search
7. Cochrane Database
8. Science Direct
9. Direct searches of relevant journals using their own search functions
10. Preprint servers

Study Selection

The title of each study identified by search will then be screened independently by two reviewers who will omit studies which clearly do not meet inclusion criteria – completing form 'SR1 – Title Screening' (Appendix 2). Any disagreements between these reviewers will prompt consideration by a third (senior) reviewer, who will arbitrate and seek consensus. If consensus is impossible, the third reviewer holds the casting vote.

Following this stage, all remaining studies will then be screened based on their abstracts following the same process as for their titles – completing form 'SR2 – Abstract Screening' (Appendix 3).

Following this, the remaining studies will then be independently read in full by two reviewers, and final decisions made regarding which studies to be included/excluded as described below.

Data Extraction and Synthesis

Two researchers will independently extract data from the studies using a pre-specified systematic review proforma – form 'SR3 – Full text Data Extraction' (Appendix 4). Once each reviewer has completed their proforma these will be compared. If in agreement, this data will be inputted into the review database. If there are disagreements between the two reviewers about one or more included data sources a third (senior) reviewer will seek consensus, and if necessary decide which data are inserted into the review database.

Where numerical data are collected, heterogeneity will be calculated using the I^2 statistical measure. Where appropriate, summary statistics will be generated to allow quantitative analysis and appropriate statistical comparisons to be performed. If the level of heterogeneity is too great, or numerical results are either incomparable or absent, a narrative synthesis will be carried out.

Risk of Bias Assessment

The risk of bias in the studies will be assessed using the Cochrane collaboration risk of bias tool in the case of RCTs, (15) and a critical appraisal will be carried out using the relevant CASP tools for all study types. (14) A study's risk of bias will be recorded in their 'SR3' form. Risk of bias will thus be

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assessed by two reviewers, with a third (senior) reviewer holding a casting vote if consensus is not agreed. Studies with severe methodological flaws will be excluded from the final review if these flaws fundamentally undermine their results.

Strength of evidence assessment

The strength of the body of evidence will be assessed and reported using the GRADE (Grading of Recommendations, Assessment, Development and Evaluations) framework (16)

Ethical review and Dissemination

Formal ethical approval for this study is not required as it is a systematic review using primary literature widely available within the public domain. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRIMSA) (17) format will be used when reporting the findings of this review. Following completion, results will be presented in the form of a written manuscript which will be submitted for publication to a relevant medical journal within the field of resuscitation, emergency care, critical care or anaesthesia. Results will additionally be submitted for presentation at relevant speciality meetings. To maximise the public's access to this review's findings, any scientific report will be accompanied by a lay summary posted via social media channels, and a press release disseminated to national and international news agencies.

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Author Statement

This protocol has been written primarily by BTS, with review and supervision conducted by JEA, NC and AO.

Patient and Public Involvement Statement

There was no patient or public involvement at the protocol stage of this project. However, all authors have experience of teaching Basic Life Support through Resuscitation for Medical Discipline (RMD) Birmingham.

Funding statement

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests statement

No competing interests were identified by any authors or reviewers.

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Appendix 1: Search Strategy

No.	Keywords [TIAB]	MeSH terms
1	(Basic Life Support **OR** BLS **OR** Basic Cardiac Life Support **OR** Life Support **OR** Resuscitation **OR** CPR **OR** Cardio Pulmonary Resuscitation) **OR** (Chest Compressions **OR** Compressions) **OR** (Rescue Breath* **OR** Mouth-to-mouth **OR** Expired Air Resuscitation **OR OR** Expired Air Ventilation **OR** EAV) **OR (**AED **OR** (Automated External) Defibrillator **OR** Defibrillation)	(Cardiopulmonary Resuscitation **OR** Heart Massage **OR** Respiration, Artificial **OR** Defibrillators)
2	Decay **OR OR** Loss **OR** Degener* **OR** Deterior* **OR** Impair* **OR** Atrophy **OR** Decline **OR** Decreas* **OR** Depreciat* **OR** Fade **OR** Fading **OR** Faded **OR** Reced* **OR** Rever* **OR** Relaps* **OR** Reduc* **OR** Wors*	**n/a**
3	Attain* **OR** Retention **OR** Retain* **OR** Increas*	**n/a**
4	Refresh* **OR** Retrain* **OR** Boost **OR** Update **OR** Renew **OR** Revise **OR** Reteach* **OR** Redo	**n/a**

To create search:

1. Combine Keyword [TIAB] and MeSH terms for each row using OR, and encapsulate in brackets ()
2. Combine each entire row with each other row using AND
3. Apply limits

Limits: None

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Appendix 2: SR1 – Title Screening Form

Tackling skill decay following Basic Life Support training – a systematic review

Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does the study title suggest it may meet the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N
	Assesses BLS skill retention via any means: Y/N
English language copy available?	Yes
	No – refer to project supervisor to assess if commissioned translation is feasible
Comments / other information:	
Decision: (at this stage, only exclude reports which do not meet the inclusion criteria. Quality of reports will be assessed later)	**Include / Exclude**

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Appendix 3: SR2 – Abstract Screening Form

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Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does the study abstract suggest it may meet the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N
	Assesses BLS skill retention via any means: Y/N
English language copy available?	Yes
	No – refer to project supervisor to assess if commissioned translation is feasible
Comments / other information:	
Decision: (at this stage, only exclude reports which do not meet the inclusion criteria. Quality of reports will be assessed later)	**Include / Exclude**

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Appendix 4: SR3 – Full-text Screening & Data Extraction Form

Tackling skill decay following Basic Life Support training – a systematic review

1. Full-text screening

Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does full-text review of the study indicate it meets the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N Assesses BLS skill retention via any means: Y/N
Comments / other information:	
Decision: At this stage, only exclude reports which obviously do not meet the inclusion criteria. Quality of reports will be assessed later. If excluding, do not complete data extraction section (to follow).	**Include / Exclude**

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2. Quality of study assessment

Complete relevant CASP checklist for study type https://casp-uk.net/casp-tools-checklists/	Copy/paste questions and responses
RCT – complete Cochrane 'Risk of Bias' tool https://www.riskofbias.info/welcome/rob-2-0-tool/current-version-of-rob-2	Copy/paste questions and responses
Assess risk of bias, based on CASP checklist +/- Cochrane tool	**Low / high**
Is there methodological flaw so severe that the study should be omitted from review?	**Yes, omit / No, do not omit – studies flagged for omission at this stage will be referred to senior review**

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3. Data extraction

a. Background

Study Title	
Study Reference (Vancouver)	
Country where study conducted List all if multiple	

b. Methods

Study design	Randomised Control Trial
	Non-randomised interventional trial
	Observational (cohort/case-control)
	Commentary/Qualitative study
	Systematic review
	Literature/Narrative review
	Other
Population Identify Key characteristics of the population studied	
Intervention(s) If applicable – i.e. was refresher training delivered?	
Comparator If applicable	
Outcome measures used I.e. what did the authors state as the primary, and secondary outcomes?	
Study inclusion criteria	
Study exclusion criteria	
Study duration	
Number of participants	
Details of initial BLS training	
Time elapsed between initial BLS training and assessment of retention	
Was any retraining received by participants <u>not as part of this study?</u>	

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Eg. As part of routine education, re-certification etc.	

c. For studies where refresher training was delivered:

Was there a comparator? Eg. Placebo, another type of refresher training?	
If there was a comparator, how was allocation to groups conducted?	**Randomised / non-randomised**
If randomised, how was this done?	
Number of study arms Eg. If just one form of refresher training vs comparator, this is 2 arms.	

d. Participant information

Average age Specify statistic – mean / median	
Sex breakdown Specify % male, % female	
Loss to follow up (n, %)	
Differences at baseline Summarise any differences between the studied groups at baseline, before any measurement or intervention took place (if reported).	
Method(s) of recruitment to study Describe / state	
Other characteristics lay-people/healthcare workers, previous BLS experience etc	

e. Participant groups and their outcomes - complete one copy of table e. for each group if multiple interventions. Also complete a copy for the control / comparator group

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Group name	
Description of group, including intervention What was done differently to this group than other participants?	
Number (%) allocated to group	
Method of BLS re-assessment	
Time between training and re-assessment	
Outcomes: where possible use raw numbers in the following format: n(%)	**CPR rate** **CPR depth** **CPR hand position** **CPR hands-off time** **Rescue breath success** **BLS algorithm completeness** **BLS summative assessment – 'pass' or 'fail'**
Other outcomes Give details of other outcome measures reported in the study. Where possible use raw numbers in the following format: n(%)	

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9

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Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	9
Sponsor	#5b	Provide name for the review funder and / or sponsor	9
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	9
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	5
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6,7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Study records - data	#11c	Describe planned method of extracting data from reports (such as	7

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7,8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7,8
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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BMJ Open

Skill decay following Basic Life Support training – a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-051959.R1
Article Type:	Protocol
Date Submitted by the Author:	16-Sep-2021
Complete List of Authors:	Stanley, Benjamin; University of Birmingham College of Medical and Dental Sciences Burton, Thomas; University of Birmingham College of Medical and Dental Sciences Percival, Harriet; University of Birmingham College of Medical and Dental Sciences Beesley, Emily; University of Birmingham College of Medical and Dental Sciences Coffin, Nicholas; University of Birmingham College of Medical and Dental Sciences; Salford Royal NHS Foundation Trust Hulme, Jonathan; University of Birmingham College of Medical and Dental Sciences; Sandwell and West Birmingham Hospitals NHS Trust Owen, Andrew; University of Birmingham College of Medical and Dental Sciences; University Hospitals Birmingham NHS Foundation Trust Alderman, Joseph; University of Birmingham College of Medical and Dental Sciences; University Hospitals Birmingham NHS Foundation Trust
Primary Subject Heading:	Medical education and training
Secondary Subject Heading:	Evidence based practice
Keywords:	MEDICAL EDUCATION & TRAINING, Adult intensive & critical care < ANAESTHETICS, PUBLIC HEALTH

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Skill decay following Basic Life Support training – a systematic review protocol

Word count: 2053

Stanley BT[1], Burton T[1], Percival H[1], Beesley E[1], Coffin N[1,2], Hulme J[1,3], Owen A[1,4], Alderman J[1,4]

1. College of Medical and Dental Sciences, University of Birmingham, UK

2. Department of Anaesthesia and Intensive Care, Salford Royal NHS Foundation Trust

3. Department of Anaesthesia and Intensive Care Medicine, Sandwell and West Birmingham NHS Trust

4. Department of Anaesthesia and Intensive Care Medicine, University Hospitals Birmingham NHS Foundation Trust, Birmingham, UK.

Corresponding author:

Joseph Alderman

joseph.alderman@nhs.net

University Department of Anaesthesia & Critical Care

Office 1, Ground Floor, East Block

Heritage Building (Queen Elizabeth Hospital)

Mindelsohn Way

Edgbaston

Birmingham

B15 2TH

Reviewing team

Name	Affiliation	Contact details	Role & tasks
Benjamin Stanley	5th year medical student. University of Birmingham (currently intercalating)	bts692@student.bham.ac.uk	Primary reviewer Student lead Project design, , project delivery, manuscript authorship
Thomas Burton	5th year medical student. University of Birmingham	txb619@student.bham.ac.uk	Reviewer Project delivery, manuscript editing and review

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	(currently intercalating)		
Harriet Percival	5th year medical student. University of Birmingham	hxp644@student.bham.ac.uk	Reviewer Project delivery, manuscript editing and review
Emily Beesley	4th year medical student. University of Birmingham (currently intercalating)	elb712@student.bham.ac.uk	Reviewer Project delivery, manuscript editing and review
Nicholas Coffin	Clinical Fellow in Intensive Care Medicine Salford Royal NHS Foundation Trust	nicholasjames.coffin@nhs.net	Project design, project delivery, manuscript editing and review, project supervision
Jonathan Hulme	Consultant, Intensive Care Medicine & Anaesthesia Sandwell and West Birmingham NHS Trust	jonathan.hulme@nhs.net	Project supervision, manuscript editing and review
Andrew Owen	NIHR Academic Clinical Lecturer in Intensive Care Medicine. University of Birmingham	andrew.owen@doctors.org.uk	Senior project supervisor Project design, project delivery, manuscript authorship, project supervision
Joseph Alderman	Anaesthesia & Intensive Care Medicine StR University Hospitals Birmingham NHS Foundation Trust	joseph.alderman@nhs.net	Tie-breaker reviewer Junior project supervisor Project design, , project delivery, manuscript authorship, project supervision

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Title: Skill decay following Basic Life Support training – a systematic review protocol

Keywords:

- Basic Life Support
- Out of Hospital Cardiac Arrest (OHCA)
- Skill Decay
- Refresher Training
- Medical Education

Abstract

Introduction

Survival from Out of Hospital Cardiac Arrest (OHCA) is lower in the United Kingdom (UK) than in several developed nations. Bystander Cardiopulmonary Resuscitation (CPR) is associated with increased rates of survival to hospital discharge following OHCA, prompting the introduction of several initiatives by the UK government to increase rates of bystander CPR, including the inclusion of Basic Life Support (BLS) teaching within the English national curriculum. Whilst there is clear benefit in this, increasing evidence suggests poor retention of skills following BLS teaching. The aim of this systematic review is to summarise the literature regarding skill decay following BLS training, reporting particularly the time period over which this occurs, and which components of would-be rescuers' performance of the BLS algorithm are most affected.

Methods and Analysis

A search will be conducted to identify studies in which individuals have received BLS training and received subsequent assessment of their skills at a later date. A search strategy comprising relevant MeSH terms and keywords has been devised with librarian input. Relevant databases will be searched with titles, abstract and full text review conducted independently by two reviewers. Data will be extracted from included studies by two reviewers, with meta-analysis conducted if the appropriate pre-conditions (such as limited heterogeneity) are met.

Ethic and Dissemination

No formal ethical approval is required for this systematic review. Results will be disseminated in the form of manuscript submission to a relevant journal and presentation at relevant meetings. To maximise the public's access to this review's findings, any scientific report will be accompanied by a lay summary posted via social media channels, and a press release disseminated to national and international news agencies.

Registration details

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This systematic review was submitted to PROSPERO on 14/02/2021 and was accepted on 17/02/2021. The PROSPERO registration number of this study is CRD42021237233

Strengths and Limitations of this study

- Support from the University of Birmingham library team has been utilised in optimisation of the search strategy, to ensure all relevant studies may be identified for inclusion
- The study methodology is robust, with clear protocol ensuring each title, abstract and full text is assessed by at least 2 reviewers
- This protocol outlines clear methodology to assess the risk of bias and the strength of evidence, using validated scoring systems, hence ensuring inclusion of high-quality primary literature
- Primary data underlying this study is likely to be heterogenous in nature, leading to challenges with conducting meta-analysis

Introduction

In the United Kingdom (UK) approximately 7-8% of patients who suffer out of hospital cardiac arrest (OHCA) survive to hospital discharge. (1) This is a considerably lower rate of survival compared with several other developed countries. (2,3) A recent report from the National Confidential Enquiry into Patient Outcome and Death (NCEPOD) demonstrates that in cases where bystander Cardiopulmonary Resuscitation (CPR) is commenced, survival to hospital discharge is increased. (4) Overall however, rates of bystander CPR remain low in the UK. (1) Increased availability of bystander CPR for patients in the UK has been set as a priority by the Department of Health to increase survival following OHCA. (5)

In order to accomplish this, the need for increased training in Basic Life Support (BLS) skill has been recognised, with BLS and first aid training a compulsory part of the English school curriculum since 2020. (6) Alongside this mandated training there have also been several national and international initiatives aimed at increasing knowledge and skills in BLS, such as the international 'Restart a Heart Day' - a joint initiative of the UK Resuscitation Council, the British Heart Foundation, British Red Cross and St John's Ambulance. (7)

The Problem:

Whilst there is undoubtedly value in increasing the number of people trained in BLS, there is growing evidence that retention of these skills is poor. Evidence from Randomised Control Trials assessing skill retention in lay populations following BLS training demonstrates a considerable decrease in BLS skills, commencing as early as 6 months following initial training (8,9) In addition there is evidence of poor skill retention in junior doctors (10), medical students (11) and nurses (12). This is despite the ability to adequately perform BLS being a requirement of the General Medical Council's (GMC) 'Outcomes for Graduates'. (13)

Need for this study:

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As outlined above, there is growing evidence demonstrating poor retention of BLS skills over time amongst both lay people and healthcare students & professionals. As yet a comprehensive systematic review of this evidence has not been conducted. There is a need for such a review to be carried out to ascertain if and how skills decay, and over what time period this occurs. Such information will facilitate future research on how to best refresh BLS skills. In the long-term, this should inform how improvements to refresher training programmes can best be implemented, with the overarching aim of improving rates of bystander CPR, and hence survival, from OHCA.

Methods and analysis

Research Questions:

- Is there evidence to support the hypothesis that Basic Life Support skills decay over time?
    - If there is evidence for skill decay, over what time period does this occur?
    - If there is evidence for skill decay, is there a difference in the way this manifests across different aspects of the BLS algorithm?
- What interventions have been implemented to address skill decay?
    - Is there evidence that these interventions are effective at addressing skill decay?

Inclusion Criteria	**Exclusion Criteria**
Study type: Systematic reviews, randomised controlled trials (RCTs), observational studies (cross-sectional, case-control or cohort), consensus statements, case series, conference papers and proceedings, other reports.	Non-English language studies where translation is not possible or cost effective
Population: Individuals who a) have received training in BLS, and b) whose skill retention has been assessed via any means. Includes: • Lay individuals • Healthcare professionals • Healthcare students	Low quality studies will be excluded from the review following assessment using Critical Appraisal Skills Programme (CASP) tool (14) and critical appraisal by at least two reviewers.
Intervention: refresher BLS training delivered via any method	Studies with severe methodological flaws such that results of the study are likely to be flawed
Outcome: Practical BLS skill performance assessed via any method	

Table 1: Inclusion and Exclusion Criteria for primary literature

Search Strategy

Published literature will be found through appropriate medical data sources, searched using relevant MeSH terms. Experienced professional colleagues within the University of Birmingham library

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services team have been consulted to optimise search terms and execution of the search, allowing all applicable articles and abstracts to be identified. The final search strategy is attached as Appendix 1. This has been piloted and found to function effectively. If required this may be altered in a minor way to fix and issues concerning syntax, spelling or grammar if discovered at a later date, or as needed to meet the requirements of each database searched. Any changes to the final search strategy will be outlined in the manuscript of the review itself.

The data sources to be searched will be OVID Medline, EMBASE, medRxiv, CINAHL, Google Scholar, Nice evidence search, the Cochrane Database and Science Direct. Reference lists of included articles will be analysed to allow any further potentially relevant articles to be foundThese will then be assessed according to the inclusion and exclusion criteria (Table 1) and included in the review if appropriate. Specific searches of the journals 'Resuscitation' and the 'Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine' will be performed given their importance in the field. Additionally, experts within the field of resuscitation research will be contacted regarding any unpublished work which may be relevant. If so, permission will be sought to access this so it may be included where appropriate. Finally, clinicaltrials.gov will be searched to identify any potentially relevant ongoing work.

Study Selection

Records of studies will be managed using Zotero software. The title of each study identified by search will then be screened independently by two reviewers who will omit studies which clearly do not meet inclusion criteria – completing form 'SR1 – Title Screening' (Appendix 2). Any disagreements between these reviewers will prompt consideration by a third (senior) reviewer, who will arbitrate and seek consensus. If consensus is impossible, the third reviewer holds the casting vote.

Following this stage, all remaining studies will then be screened based on their abstracts following the same process as for their titles – completing form 'SR2 – Abstract Screening' (Appendix 3).

Following this, the remaining studies will then be independently read in full by two reviewers, and final decisions made regarding which studies to be included/excluded as described below.

Data Extraction and Synthesis

A bespoke database will be produced specifically for this project, using commercially available database software. Two researchers will independently extract data from the studies using a pre-specified systematic review proforma – form 'SR3 – Full text Data Extraction' (Appendix 4). This data extraction form has been piloted and found to work effectively. Once each reviewer has completed their proforma these will be compared. If in agreement, this data will be inputted into the review database. If there are disagreements between the two reviewers about one or more included data sources a third (senior) reviewer will seek consensus, and if necessary decide which data are inserted into the review database.

Where numerical data are collected, heterogeneity will be calculated using the I^2 statistical measure. Where appropriate, summary statistics will be generated to allow quantitative analysis and appropriate statistical comparisons to be performed. Statistical analysis will be performed using 'R'

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programming. (15) If the level of heterogeneity is too great, or numerical results are either incomparable or absent, a narrative synthesis will be carried out.

Risk of Bias Assessment

The risk of bias in the studies will be assessed using the Cochrane collaboration risk of bias tool in the case of RCTs, (16) and a critical appraisal will be carried out using the relevant CASP tools for all study types. (14) A study's risk of bias will be recorded in their 'SR3' form. Risk of bias will thus be assessed by two reviewers, with a third (senior) reviewer holding a casting vote if consensus is not agreed. Studies with severe methodological flaws will be excluded from the final review if these flaws fundamentally undermine their results.

Strength of evidence assessment

The strength of the body of evidence will be assessed and reported using the GRADE (Grading of Recommendations, Assessment, Development and Evaluations) framework (17)

Ethical review and Dissemination

Formal ethical approval for this study is not required as it is a systematic review using primary literature widely available within the public domain. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRIMSA) (18) format will be used when reporting the findings of this review. Following completion, results will be presented in the form of a written manuscript which will be submitted for publication to a relevant medical journal within the field of resuscitation, emergency care, critical care or anaesthesia. Results will additionally be submitted for presentation at relevant speciality meetings. To maximise the public's access to this review's findings, any scientific report will be accompanied by a lay summary posted via social media channels, and a press release disseminated to national and international news agencies.

Discussion

Whilst there has been a considerable increase in the availability of Basic Life Support teaching in recent years, it remains unclear how the skills acquired decay over time. This systematic review therefore aims to address this. The results from this review will provide clearer guidance on the optimal management of skill decay following BLS teaching, therefore aiding the design effective 'refresher' training programmes in continuous professional development.

The study protocol is robust, and identifies a clear structure of study selection, data extraction and assessment of risk of bias and strength of evidence. Potential limitations which should be acknowledged include lack of identifiable studies, however initial scoping searches suggest sufficient literature to warrant the undertaking of a systematic review. Additionally, the complexity of the research question may limit analysis, as may the range of study types included. A further consideration is the potential for a high level of heterogeneity between included studies, which may limit any meta-analyses. This protocol accounts for this in that a narrative synthesis will be performed in this case, alongside any appropriate sub-group analyses.

In conclusion, whilst this review does have potential limitations, this is outweighed by the importance of the research question, and the potential for improvements in BLS education the review may therefore have.

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Author Statement

The project was conceived, designed and initialised by BS, JA, JH, AO, NC, EB, TB and HP. The protocol manuscript was initially drafted by BS, with supervision from JH, AO, JA and NC and then reviewed and amended by all authors. Final re-drafting was completed by BS, AO, JA, JH and NC and approved by all authors. Adaptations and response to peer-review was completed by BS, with supervision from AO and JA. All authors have approved the final submission and are accountable for its accuracy.

Patient and Public Involvement Statement

There was no patient or public involvement at the protocol stage of this project. However, all authors have experience of teaching Basic Life Support through Resuscitation for Medical Discipline (RMD) Birmingham.

Funding statement

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests statement

No competing interests were identified by any authors or reviewers.

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Appendix 1: Search Strategy

No.	Keywords [TIAB]	MeSH terms
1	(Basic Life Support **OR** BLS **OR** Basic Cardiac Life Support **OR** Life Support **OR** Resuscitation **OR** CPR **OR** Cardio Pulmonary Resuscitation) **OR** (Chest Compressions **OR** Compressions) **OR** (Rescue Breath* **OR** Mouth-to-mouth **OR** Expired Air Resuscitation **OR** Expired Air Ventilation **OR** EAV) **OR (**AED **OR** (Automated External) Defibrillator **OR** Defibrillation)	(Cardiopulmonary Resuscitation **OR** Heart Massage **OR** Respiration, Artificial **OR** Defibrillators)
2	Decay**OR** Loss **OR** Degener* **OR** Deterior* **OR** Impair* **OR** Atrophy **OR** Decline **OR** Decreas* **OR** Depreciat* **OR** Fade **OR** Fading **OR** Faded **OR** Reced* **OR** Rever* **OR** Relaps* **OR** Reduc* **OR** Wors*	**n/a**
3	Attain* **OR** Retention **OR** Retain* **OR** Increas*	**n/a**
4	Refresh* **OR** Retrain* **OR** Boost **OR** Update **OR** Renew **OR** Revise **OR** Reteach* **OR** Redo	**n/a**

To create search:

1. Combine Keyword [TIAB] and MeSH terms for each row using **OR**, and encapsulate in brackets ()
2. Combine rows 2, 3 and 4 using **OR**
3. Combine this search result with row 1 using **AND**
4. Apply limits

Limits: None

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Appendix 2: SR1 – Title Screening Form

Tackling skill decay following Basic Life Support training – a systematic review

Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does the study title suggest it may meet the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N
	Assesses BLS skill retention via any means: Y/N
English language copy available?	Yes
	No – refer to project supervisor to assess if commissioned translation is feasible
Comments / other information:	
Decision: (at this stage, only exclude reports which do not meet the inclusion criteria. Quality of reports will be assessed later)	**Include / Exclude**

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Appendix 3: SR2 – Abstract Screening Form

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Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does the study abstract suggest it may meet the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N
	Assesses BLS skill retention via any means: Y/N
English language copy available?	Yes
	No – refer to project supervisor to assess if commissioned translation is feasible
Comments / other information:	
Decision: (at this stage, only exclude reports which do not meet the inclusion criteria. Quality of reports will be assessed later)	**Include / Exclude**

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Appendix 4: SR3 – Full-text Screening & Data Extraction Form

Tackling skill decay following Basic Life Support training – a systematic review

1. Full-text screening

Date of screening	
Reviewer	Electronic form will have dropdown list of reviewer names
Study details Title, URL, citation	
Any conflicts of interest identified by the reviewer? Eg. Were they involved in the publication?	
Does full-text review of the study indicate it meets the INCLUSION criteria:	Involves individuals who have received BLS training: Y/N Assesses BLS skill retention via any means: Y/N
Comments / other information:	
Decision: At this stage, only exclude reports which obviously do not meet the inclusion criteria. Quality of reports will be assessed later. If excluding, do not complete data extraction section (to follow).	**Include / Exclude**

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2. Quality of study assessment

Complete relevant CASP checklist for study type https://casp-uk.net/casp-tools-checklists/	Copy/paste questions and responses
RCT – complete Cochrane 'Risk of Bias' tool https://www.riskofbias.info/welcome/rob-2-0-tool/current-version-of-rob-2	Copy/paste questions and responses
Assess risk of bias, based on CASP checklist +/- Cochrane tool	**Low / high**
Is there methodological flaw so severe that the study should be omitted from review?	**Yes, omit / No, do not omit – studies flagged for omission at this stage will be referred to senior review**

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3. Data extraction

a. Background

Study Title	
Study Reference (Vancouver)	
Country where study conducted List all if multiple	

b. Methods

Study design	Randomised Control Trial
	Non-randomised interventional trial
	Observational (cohort/case-control)
	Commentary/Qualitative study
	Systematic review
	Literature/Narrative review
	Other
Population Identify Key characteristics of the population studied	
Intervention(s) If applicable – i.e. was refresher training delivered?	
Comparator If applicable	
Outcome measures used I.e. what did the authors state as the primary, and secondary outcomes?	
Study inclusion criteria	
Study exclusion criteria	
Study duration	
Number of participants	
Details of initial BLS training	
Time elapsed between initial BLS training and assessment of retention	
Was any retraining received by participants <u>not as part of this study?</u>	

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Eg. As part of routine education, re-certification etc.	

c. For studies where refresher training was delivered:

Was there a comparator? Eg. Placebo, another type of refresher training?	
If there was a comparator, how was allocation to groups conducted?	**Randomised / non-randomised**
If randomised, how was this done?	
Number of study arms Eg. If just one form of refresher training vs comparator, this is 2 arms.	

d. Participant information

Average age Specify statistic – mean / median	
Sex breakdown Specify % male, % female	
Loss to follow up (n, %)	
Differences at baseline Summarise any differences between the studied groups at baseline, before any measurement or intervention took place (if reported).	
Method(s) of recruitment to study Describe / state	
Other characteristics lay-people/healthcare workers, previous BLS experience etc	

e. Participant groups and their outcomes - complete one copy of table e. for each group if multiple interventions. Also complete a copy for the control / comparator group

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Group name	
Description of group, including intervention What was done differently to this group than other participants?	
Number (%) allocated to group	
Method of BLS re-assessment Include any information concerning training of BLS assessors	
Time between training and re-assessment	
Outcomes: where possible use raw numbers in the following format: n(%)	**CPR rate** **CPR depth** **CPR hand position** **CPR hands-off time** **Rescue breath success** **BLS algorithm completeness** **BLS summative assessment – 'pass' or 'fail'**
Other outcomes Give details of other outcome measures reported in the study. Where possible use raw numbers in the following format: n(%)	

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	9

Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	9
Sponsor	#5b	Provide name for the review funder and / or sponsor	9
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	9
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	5
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6,7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	10
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7
Study records - data	#11c	Describe planned method of extracting data from reports (such as	7

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collection process		piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7,8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	7
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	7
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	7
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	7,8
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	8

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